ENGINEERS IN ACTION

Project Name: NYAKUZA SOSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: SIMISO DLAMINI
Construction Manager/Foreman: MICHAEL MNISI

6.1 - CONSTRUCTION LAYOUT QUALITY CONTROL FORM

Initial to confirm each step has been completed. Boxes filled with gray are not required to be initialed. indicates QC photo required. indicates as-built required.

Step	Description	Construction Manager Initials	Technical Supervisor Initials
Step 1	Mark survey points with spray paint or another defined marker.	[signature]	
	Establish the centerline of the bridge across the river with a string line or surveying equipment.		SD
Step 2	Establish left and right areas of construction and clear the areas of debris obstructing layout.	[signature]	
	Take photos of bridge centerline from upstream (A1) and downstream (A2).	[signature]	
	Verify bridge span length.		SD
	Place stake along centerline at the front of left foundation.	[signature]	
	Place stakes along centerline at the front of right foundation.	[signature]	
Step 3	Place stake along centerline at the back of left anchor.	[signature]	
	Place stake along centerline at the back of right anchor.	[signature]	
	Place construction benchmark along centerline in front of and behind left abutment. Benchmarks should be placed outside of excavation area. Record distance along centerline from each benchmark to saddle.	[signature]	
	Place construction benchmark along centerline in front of and behind right abutment. Benchmarks should be placed outside of excavation area. Record distance along centerline from each benchmark to saddle.	[signature]	
Step 4	Place stakes at each corner of left foundation and anchor and confirm square to centerline.	[signature]	
	Place stakes at each corner of right foundation and anchor and confirm square to centerline.	[signature]	
	Mark with paint/chalk excavation area and centerline on ground on both left side (A3L) and right side (A3R).	[signature]	
	Verify Quality Control photos have been taken.	[signature]	

Project Name: NYAKUZA ~~[illegible]~~ SUSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: JEREMY RYDING
~~Construction~~ QC Manager/Foreman: MADELINE STEVENS, SIENNA GIUSEPPI

6.2 - EXCAVATION QUALITY CONTROL FORM

Initial to confirm each step has been completed. (camera icon) indicates QC photo required. (pencil icon) indicates as-built required.

Step	Description	Construction Manager Initials: Left Abutment	Construction Manager Initials: Right Abutment
Step 1	Excavate foundation to elevation shown on drawings and compact base material (B1L / B1R).	[signature]	[signature]
	If water is seeping into excavations, provide means of drainage.	–	–
	Record as-built bottom of foundation elevations.	[signature]	[signature]
Step 2	Excavate anchor to elevation shown on drawings and compact base material (E1L / E1R).	MS	~~MS~~ SG
	If water is seeping into excavations, provide means of drainage.	–	–
	Record as-built bottom of anchor elevations.	MS	~~MS~~ SG
Step 3	Excavate soil for ramp walls (G1L / G1R).	MS	SG
Step 4	Excavate soil between ramp walls for cable placement.	MS	SG
			SG
	Verify Quality Control photos have been taken.	MS	SG

6.2 - EXCAVATION SAFETY

Description	Technical Supervisor Initials
Verify all excavations are properly benched or sloped. Benching required for excavations deeper than 1.5 meters. Lowest bench not to exceed a height of 1.5 meters above bottom of excavation. Subsequent benches must have a maximum height of 1 meter and minimum width of 1 meter.	
Verify all excavated soil is piled at least one meter back from edge of excavations.	

ENGINEERS IN ACTION

Project Name: NYAKUZA SUSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: JEREMY RYDING
QC ~~**Construction**~~ **Manager/Foreman:** MADELINE STEVENS, SIENNA GIUSEPPI

6.3 – FOUNDATION AND TIER QUALITY CONTROL FORM

Initial to confirm each step has been completed. Boxes filled with gray are not required to be initialed. [camera icon] indicates QC photo required. [pencil icon] indicates as-built required.

	Step	Description	Construction Manager Initials							
			Left Abutment				Right Abutment			
			F	1	2	3	F	1	2	3
	Step 1	Verify no water is seeping into excavation. Provide drainage if present.								
		Verify foundation excavations are level, clear of debris, and square to centerline.	[illegible]							
	Step 2	Place stones at each corner of foundation/tier.	[illegible]	SG	MS			MS	SG	
[camera]	Step 3	Construct masonry walls square and plumb (B2L / B2R / C1L / C1R). Verify wall ✓ thickness.	[illegible]	MS	SG			MS	SG	
	Step 4	Use spray paint to mark the inside of the masonry walls indicating the heights of each 30cm lift for the interior rock fill.	[illegible]	MS	SG			MS	SG	
[camera]		Fill interior of foundation/tier with rocks in 30cm lifts. On the top of each lift, fill voids between rocks with smaller rocks and cover with a layer of concrete slurry. Fill must not exceed 3 lifts per day. Do not use soil for fill. (B3L / B3R / C2L / C2R).	[illegible]	MS	SG			MS	SG	
[camera]	Step 5	Cast perimeter topping slab on top of the outer edges of each foundation (B4L / B4R) and tier (C3L / C3R). The slab shall be 70 centimeters wide, a minimum of five centimeters thick, and must be roughened wherever it will carry the next level of masonry.	MS	MS	SG			MS	SG	
[pencil]		Record as-built length, width, height, and elevation of foundation/tier.	SG	MS	MS			MS	SG	
		Verify Quality Control photos have been taken.	SG	MS	MS			MS	SG	

ENGINEERS IN ACTION

6.4 – TOWER QUALITY CONTROL FORM

Project Name: NYAKUZA SUSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: DAVID SCHAAD
QC ~~**Construction**~~ **Manager/Foreman:** MADELINE STEVENS, SIENNA GIUSEPPI

Initial to confirm each step has been completed. Boxes filled with grey are not required to be initialed. (camera icon) indicates QC photo required (pencil icon) indicates as-built required.

Step	Description	Construction Manager Initials – Left Abutment	Construction Manager Initials – Right Abutment	Technical Supervisor Initials
Step 1	Verify span length and difference in elevation at top of tiers.			DES
	Mark tower base dimensions square to centerline.	MS	SG	
Step 2	Bend reinforcing bars per tower drawings.	MS	SG	
Step 3	Construct 20cm brick base and place reinforcing cage with 7.5cm of clearance (D1L / D1R).	MS	SG	
	Fill brick base with concrete, embedding lower part of reinforcing cage (D2L / D2R).	MS	SG	
Step 4	Construct first 120cm of towers ensuring 7.5cm of clearance for reinforcing cage (D3L / D3R).	MS	SG	
	Fill brick tower with concrete in 20-40 cm lifts. Leave clearance for steel wheel (D4L / D4R).	MS	SG	
Step 5	Install handrail cable saddles (steel wheels) such that cable bearing points are 150cm above top of tier and level to one another (D5L / D5R).	MS	SG	
Step 6	Complete top of towers with concrete so that cable bearing point remains exposed (D6L / D6R).	MS	SG	
Step 7	Construct walkway hump to a height 110cm below cable bearing point on handrail cable saddles. Embed cable guide bars (D7L / D7R).	MS	SG	
	Record as-built elevation of walkway humps above a point of known elevation tied into the design. Calculate and record the as-built elevation of the walkway humps above the High Water Line (HWL).			DES
	Record as–built ΔH by measuring elevation difference between the walkway humps.			DES
	Record as-built horizontal distance from bridge centerline to center of handrail saddles and center of walkway cable guides.			DES
	Record as-built vertical distance between the handrail cable bearing point (groove in steel wheel) and the walkway cable bearing point (top of the walkway hump).			DES
	Verify Quality Control photos have been taken.	MS	MS	

ENGINEERS IN ACTION

Project Name: NYAKUZA SUSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: MELANIE CEDENO, [redacted] CHRIS BUSCH, DAVID SCHAAD
QC ~~Construction~~ Manager/Foreman: MADELINE STEVENS, SIENNA GIUSEPPI

6.5 – ANCHOR AND CABLE PREPARATION QUALITY CONTROL FORM

Initial to confirm each step has been completed. Boxes filled with grey are not required to be initialed. (camera icon) indicates QC photo required. (pencil icon) indicates as-built required.

Step	Description	Construction Manager Initials: Left Abutment	Construction Manager Initials: Right Abutment	Technical Supervisor Initials
Step 1	Verify no water is seeping into excavation. Provide drainage if present.	—	—	
	Bend anchor beam reinforcing bars and assemble cages. Wrap suction tubes around anchor cage and verify tubes are properly spaced as shown in drawings.	MS	MS	
	Place anchor reinforcing cage in excavation with proper orientation as specified in Bridge Binder (E2L / E2R). Verify erection hooks are installed in the reinforcing cage of the adjustable anchor.	MS	MS	
Step 2	Cast anchor beams (E3L / E3R).	MS	MS	
	Verify anchor has cured properly. Record as-built length, width, and height of anchor beams.	MS	MS	
Step 3	Unspool cables, inspect for damages or welds and verify diameter (F1).			M.C.
	Verify with the Program Manager that all clamps are drop-forged.			M.C.
Step 4	On the side with the fixed anchor, sleeve walkway cables through tubes at walkway hump and drape handrail cables over towers.	MS	—	
	Thread cables through tubes at fixed anchor and install all of the clamps. The quantity, spacing and size of the clamps is based on the cable diameter and specified in the Bridge Binder. The clamp saddle shall lie against the live end of the cable (never saddle a dead horse). The U-bolt of the clamp should lie against the dead end of the cable. Record as-built number of clamps per cable and spacing between clamps at fixed anchor (F2L or F2R).			OB (Left side) DES (right)
	Fully tighten the cable clamps at fixed anchor per manufacturer's specified torque (or use table in Bridge Binder), or until a 25% reduction in cross section of the dead end of the cable has been achieved (F3L or F3R).	MS	MS	
Step 5	On side with adjustable anchor, sleeve walkway cables through tubes at walkway hump and drape handrail cables over towers.	—	MS	
	Thread handrail and walkway cables through tubes at adjustable anchor.	—	MS	
Step 6	Hand hoist cables as high as possible [stay below construction sag (h_1)] on adjustable side. Install two clamps on each cable. Tighten the cable clamps until visible reduction in the cross section of the dead end of the cable has been achieved (see Bridge Binder for more details).			DES
	Verify Quality Control photos have been taken.	SG	SG	

ENGINEERS IN ACTION

Project Name: NYAKUZA SOSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: –
QC **Construction Manager/Foreman:** Madeline Stevens, Sienna Guseppi

6.6 – APPROACH RAMP (STAGE 1) QUALITY CONTROL FORM

Initial to confirm each step has been completed. indicates QC photo required. indicates as-built required.

Step	Description	Construction Manager Initials	
		Left Abutment	Right Abutment
Stage 6.6	Verify excavation is free of debris and standing or seeping water.	SG	MS
	Verify wall thickness and outside-to-outside ramp width at base of walls.	SG	MS
	Construct walls connecting the foundation to the anchor beam up to a height equal to the top of the foundation (G2L / G2R).	~~SG~~ SG	MS
	Record as-built dimensions of ramp walls.	SG	MS
	Verify Quality Control photos have been taken.	SG	MS

ENGINEERS IN ACTION

Project Name: NYAKUZA SUSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: CHRIS BUSCH, DAVID SCHAAD
QC ~~Construction~~ Manager/Foreman: MADELINE STEVENS, SIENNA GIUSEPPI

6.7 – CABLE HOISTING QUALITY CONTROL FORM

Initial to confirm each step has been completed. Boxes filled with grey are not required to be initialed. 📷 indicates QC photo required. ✏ indicates as-built required.

Step	Description	Construction Manager Initials	Technical Supervisor Initials
✏ Step 1	Recalculate and record both f values using as-built elevation difference (ΔH) and as-built span (L). Use construction sag (h_1) and hoisting sag (h_2) referenced from drawings.		DES
	Mark f_1 and f_2 for both handrail and walkway cables as per the instructions in the Bridge Binder (there should be a total of four well-labeled marks).		CB
Step 2	Place the automatic level away from danger zones such that f marks and the location of the cable low point can be seen.		CB
Step 3	Use hoisting loop to connect winch to main cable. DO NOT attach winch throw directly to main cable. Ensure cables are not crossing one another prior to hoisting.		CB
Step 4	Use winch to hoist handrail and walkway cables to construction sag (h_1). Wait a minimum of 24 hours to relieve construction stretch (see Bridge Binder).		CB
Step 5	Lower and adjust cables to hoisting sag (h_2). Verify handrail and walkway cables are level relative to each other. The handrail and walkway cables should be 110cm apart vertically.		DES
✏ 📷	Install clamps on each cable at adjustable anchor. The quantity, spacing and size of the clamps is based on the cable diameter and specified in the Bridge Binder. The clamp saddle shall lie against the live end of the cable (never saddle a dead horse). The U-bolt of the clamp should lie against the dead end of the cable. Record as-built number of clamps per cable and spacing between clamps at adjustable anchor (F2L or F2R).		DES
📷	Fully tighten the cable clamps at adjustable anchor per manufacturer's specified torque (or use table in Bridge Binder), or until a 25% reduction in cross section in the dead end of the cable has been achieved (F3L or F3R).		DES
	Verify Quality Control photos have been taken.	MS	

6.7 – CABLE HOISTING SAFETY

Description	Technical Supervisor Initials
"Winch Safety" protocol followed from Section 6.7.	DES
"Cable Safety" protocol followed from Section 6.7.	DES

ENGINEERS IN ACTION

Project Name: NYAKUZA SOSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: CHESTER WERTS, GREG ZEISS
QC ~~Construction~~ **Manager/Foreman:** MADELINE STEVENS, SIENNA GIUSEPPI

6.8 – APPROACH RAMP (STAGE 2) QUALITY CONTROL FORM

Initial to confirm each step has been completed. indicates QC photo required. indicates as-built required.

Step	Description	Construction Manager Initials	
		Left Abutment	Right Abutment
Stage 6.8	Construct 30cm thick backwall above anchor beam.	SG	MS
	Complete remaining ramp walls. Verify the thickness of the walls.	SG	MS
	Begin to fill interior of ramp with rocks in 30cm lifts. On the top of each lift, fill voids between rocks with smaller rocks and cover with a layer of concrete slurry. Fill must not exceed 3 lifts per day. Do not use soil for fill. Leave both cables and clamps fully exposed in case adjustment is necessary after installation of decking.	SG	MS
	Record as-built dimensions of approach ramp and ramp walls.	SG	MS
	Verify Quality Control photos have been taken.	SG	MS

ENGINEERS IN ACTION

Project Name: NYAKUZA SUSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: CHESTER WERTS, GREG ZEISS
Construction Manager/Foreman: MADELINE STEVENS, SIENNA GIUSEPPI

6.9 – WALKWAY QUALITY CONTROL FORM

Initial to confirm each step has been completed. Boxes filled with grey are not required to be initialed. 📷 indicates QC photo required. ✏ indicates as-built required.

	Step	Description	Construction Manager Initials	Technical Supervisor Initials
		Fall protection protocol followed Bridge Binder.		GZ
	Step 1	Verify crossbeam, nailer, decking board and suspender dimensions.	SG	
		Verify pre-drilled holes in crossbeams for suspenders are oversized with respect to suspender diameter.		GZ
	Step 2	Assemble all swings at the tower to ensure correct suspender length and launch all swings to their position before beginning decking. Install clamps on outer walkway cables on river side of first crossbeam.	SG	
📷	Step 3	Install decking boards in staggered formation with proper size and quantity of fasteners (H1).	SG	
📷	Step 4	Measure level of the deck at midspan. Make adjustments to level deck if tilted (H2).		CW
		Verify spacing between handrail and walkway cables is 110cm.		CW
📷		Verify suspender/crossbeam connection (H3).	MS	
📷		Verify suspenders tightly wrapped around handrail cables (H4).	MS	
📷	Step 5	Verify fencing installed and sufficiently fastened to both decking and handrail cable (H5).	SG	
📷	Step 6	Fill lower part of anchor tubes with grout to prevent water from entering (F4L / F4R).	MS	
		Re-torque all clamps on the fixed and adjustable anchors to manufacturer's specifications.	SG	
📷		Apply waterproof coating to portion of cables which will remain embedded in the approach ramp (F5L / F5R).	SG	
✏		Record as-built dimensions for crossbeams, nailers, decking boards, and predrilled holes.	MS	
		Verify Quality Control photos have been taken.	MS	

ENGINEERS IN ACTION

Project Name: ~~[illegible]~~ NYAMUZA SUSPENDED
Country: ESWATINI
District/Municipality: SHISELWENI
Technical Supervisor: CHESTER WERTS, GREG ZEISS
Construction Manager/Foreman: MADELINE STEVENS, SIENNA GIUSEPPI

6.10 — APPROACH RAMP (STAGE 3) AND COMPLETED BRIDGE QUALITY CONTROL FORM

Initial to confirm each step has been completed. Boxes filled with grey are not required to be initialed. (camera) indicates QC photo required. (pencil) indicates as-built required.

Step	Description	Construction Manager Initials	Technical Supervisor Initials
Step 7	Coat embedded portion of cable with roofing tar of asphalt paint.	SG	
Step 8	Complete ramp fill with rocks in 30cm lifts. On the top of each lift, fill voids between rocks with smaller rocks and cover with a layer of concrete slurry. Fill must not exceed 3 lifts per day. Do not use soil for fill. For the final lift (top of ramp) fill voids between rocks with crushed rock and gravel. Do not use soil for fill (G3L / G3R).	SG	
	Cast minimum 10cm concrete topping slab over ramp surface (G4L / G4R).	SG	
	After curing, cut contraction joints in concrete topping slab (see drawings for spacing and location).		
Step 9	Cast concrete topping between towers over walkway hump. Verify sufficient cover above cable tubes (G5L / G5R).	MS	
Step 10	Use automatic level to record the as-built dead load sag (h_3) of the completed bridge.		GZ
	Use spray paint to mark handrail cables at center of saddle and walkway cables at the opening of walkway hump tube.	MS	
	Take photos of completed bridge from upstream, downstream, left side and right side (I1 / I2 / I3L / I3R).	SG	
	Take banner photos (I4 / I5)	SG	
	Verify Quality Control photos have been taken.	SG	

ENGINEERS IN ACTION

Quality Control Forms Signature Page

By signing below, I acknowledge my involvement and oversight of the construction phase of this pedestrian bridge project. I understand that by initialing any of the quality control points from the attached Quality Control Forms, I am giving my professional word that the quality control items were completed safely and accurately in compliance with EIA Bridge Binder and approved modifications to the best of my knowledge. I understand that if I do not feel comfortable initialing any of the quality control points, it is my responsibility to make a plan with EIA for someone else to initial those points, and I commit to suspend the progress of associated construction activities until the alternate quality control representative is on site.

Print: Melanie Cedeno Signature:
Date: 6/26/25

Print: Chris Busch Signature: Chris Busch
Date: 7/2/25

Print: DAVID SCHAAP Signature:
Date: 7/10/25

Print: GREG TEISS Signature:
Date: 7/25/2024

Print: Chester Weeks Signature:
Date: 7/25/2025

Print: ____________ Signature: ____________
Date: ____________

Print: ____________ Signature: ____________
Date: ____________

Print: ____________ Signature: ____________
Date: ____________

method: Hand and Spade mixing

Concrete Mixing Log

Date	Mix Ratio	Location on Bridge	Slump	Cylinders (Yes/No)	Approver Initials	Cement bags
16-05-25	1:2	mixing pad RHS			T.S	5
16-05-25	1:2	mixing pad LHS			T.S	5
20-05-25	1:2	foundation wall LHS			T.S	15
22-05-25	1:2	foundation walls LHS			T.S	10
23-05-25	1:3	filling foundation LHS			T.S	8
27-05-25	1:2:3	caping LHS foundation			T.S	16
28-05-25	1:2	foundation wall RHS			T.S	15
~~30-05-25~~	~~1:2~~	~~Foundations wall RHS~~ filling foundation wall			T.S	~~13~~
30.05-25	1:2	Foundation wall RHS			T.S	8
30.05.25	1:3	filling foundation wall RHS			T.S	5
05 ~~02~~-06-25	1:2:3	RHS foundation cap			MS	17

Approval by a competent person requires witness of proper proportioning, mixing, and placement of concrete.